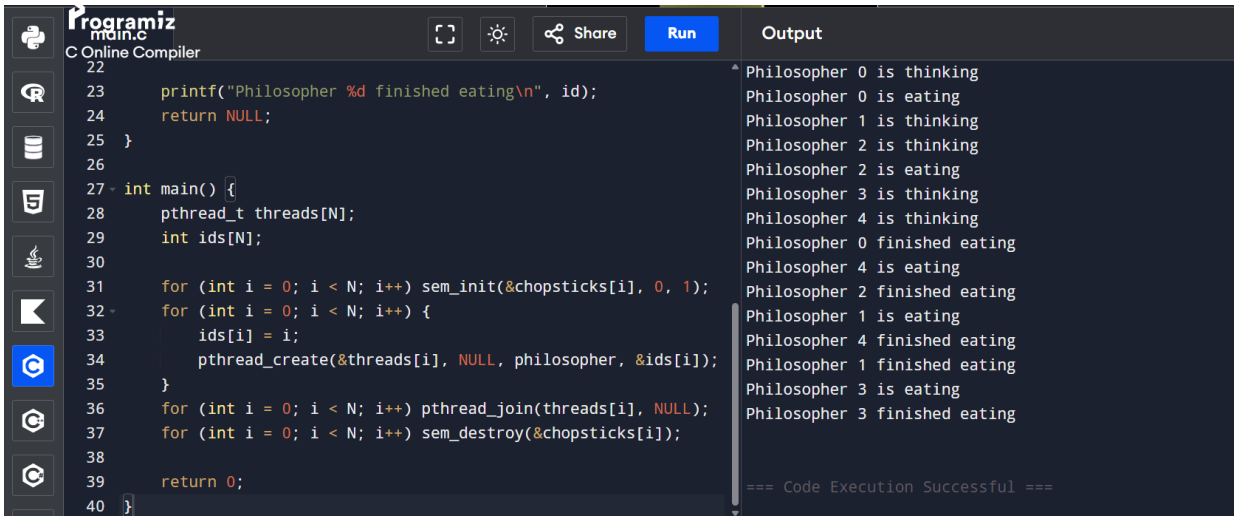
EXPERIMENT 12:

12. Design a C program to simulate the concept of Dining-Philosophers problem.



PSEUDO CODE :

1. Initialize an array of semaphores for N chopsticks.
2. Create N philosopher threads.
3. Each philosopher thread loops:
   a. Think
   b. Wait on left chopstick sem_wait(&chopstick[id])
   c. Wait on right chopstick sem_wait(&chopstick[(id+1)%N])
   d. Eat
   e. Release right and left chopstick sem_post()

CODE:

```
#include <stdio.h>
#include <pthread.h>
#include <semaphore.h>
#include <unistd.h>

#define N 5
sem_t chopsticks[N];

void* philosopher(void* num) {
    int id = *(int*)num;

    printf("Philosopher %d is thinking\n", id);

    sem_wait(&chopsticks[id]);
    sem_wait(&chopsticks[(id + 1) % N]);

    printf("Philosopher %d is eating\n", id);
    sleep(1);

    sem_post(&chopsticks[id]);
    sem_post(&chopsticks[(id + 1) % N]);

    printf("Philosopher %d finished eating\n", id);
    return NULL;
}
```

```
int main() {
    pthread_t threads[N];
    int ids[N];

    for (int i = 0; i < N; i++) sem_init(&chopsticks[i], 0, 1);
    for (int i = 0; i < N; i++) {
        ids[i] = i;
        pthread_create(&threads[i], NULL, philosopher, &ids[i]);
    }
    for (int i = 0; i < N; i++) pthread_join(threads[i], NULL);
    for (int i = 0; i < N; i++) sem_destroy(&chopsticks[i]);

    return 0;
}
```